

Suppose a Company has two VLANs:

VLAN 10 – Sales , PC1 IP : 192.168.1.10

VLAN 20 – Support, PC2 IP: 192.168.2.10

PC 1 try to ping PC 2

PC 1 > ping 192.168.2.10

Request time out.

Request time out.

Request time out.

Request time out.

Why Does the Ping Fail?

The ping fails because:

- VLAN 10 and VLAN 20 are **separate broadcast domains**.
- A **Layer 2 switch cannot route traffic** between different VLANs.
- Without a **Layer 3 device**, packets from VLAN 10 cannot reach VLAN 20.

How to Allow Inter-VLAN Communication

You need one of these:

Option 1: Router-on-a-Stick (Most common in Packet Tracer)

- Connect a router to the switch using a **trunk port**.
- Create **subinterfaces** on the router:
 - G0/0.10 for VLAN 10
 - G0/0.20 for VLAN 20
- Configure:
 - encapsulation dot1Q 10
 - encapsulation dot1Q 20
- Assign gateway IPs such as:
 - 192.168.1.10

- 192.168.2.20

The router will route traffic between VLANs.

Option 2: Layer 3 Switch

Use a Layer 3 switch and:

- Create SVIs (Switch Virtual Interfaces)
- Enable:

Command : `ip routing`

The switch itself performs the routing between VLANs.